NEER: what to run next

Status on 2026-10-01 · repository committed up to 533df7d · tests green

Where things stand

Built and committed: the six-agent loop with the AI/rules split and rule veto, the live LLM chain (Groq first, rules last), and the Control Room (city map, agent flowchart, decision cards, twin sandbox, operations board, PDF case report, light and dark theme). Rules-brain evaluation: 79/80 on seeds 41-45, 0 wrong closures, 0 unsafe actions.

Open gap: docs/LLM_EVAL.md still comes from a run where the free LLM quota ran out, so only 13 of 337 decisions were answered by a live model. The live AI has not yet been measured at scale.

1. Run the live LLM evaluation (most important)

Needs Groq quota (free tier: 200,000 tokens per day for gpt-oss-120b). Takes roughly 15 to 30 minutes.

```
NEER_LLM_MIN_INTERVAL_S=6 uv run python -m neer.eval.llm_eval --seeds 41,42,43,44,45
```

Then open docs/LLM_EVAL.md. Check: live answers are the majority, 0 wrong closures and 0 unsafe actions in both columns, and read the disagreement list. If quota ran out again, the file says so at the top; wait and re-run. When it is a real run, delete the quota warning in the file and commit.

```
git add docs/LLM_EVAL.md && git commit -m "stage8/llm: live evaluation"
```

2. Quick health checks

```
uv run pytest -q
uv run python -m neer.eval.run_eval --seeds 41,42,43,44,45
```

Refreshes docs/EVAL.md (rules brain).

```
uv run python -m neer.agents.runner --scenario leak_20 --mode mcp --llm chain
```

The last command proves the tools also work through the four MCP servers with the real LLM.

3. Only if you quote these numbers in the pitch

```
uv run python scripts/engine_report.py 40
uv run python scripts/search_bench.py 120
uv run python -m neer.eval.ablation 36
```

Project rule: quote only numbers printed by these scripts or by run_eval. Area scores are relative likelihoods, not calibrated probabilities. Do not claim live SCADA.

4. Start the demo

```
uv run streamlit run neer/app/dashboard.py
```

Suggested order (about 4 minutes):

- Operations board tab: press Run all demo scenarios. Shows a dismissed pump trip, a wrong repair caught, two leaks, and the impact strip.
- Sidebar: choose LLM chain, scenario leak_20, press Start case. Point at the badges (AI: groq vs rules) and the agent flowchart.
- Try a chaos switch (SCADA times out, broken LLM JSON) and show the case still closes FIXED.
- Twin sandbox tab: a 20 L/s leak at junction 239, Simulate, then Run a case on this.
- Operations board: Download case report (PDF). Label on every page says the city is simulated.

5. Before the final review

- Keys are in .env (ignored by git). Do not paste them anywhere. Free-tier Gemini allows only 20 requests per day, so Groq carries the demo.
- If the venue wifi is bad: the map falls back to the schematic by itself and the LLM chain falls back to rules. NEER never blocks.
- Untracked files in the repo that are not part of this work: prompts/AI_VALIDATION_PROMPTS.md, prompts/UPGRADE_PROMPTS.md, scripts/export_dataset.py. Decide whether to commit or delete them.
- Known limits to say out loud: single-click junction pick on the sandbox map is unreliable (drag a small box instead); two_leaks seed 43 escalates to a human (fails safe); all data is simulated (EPA Net3 placed on a Madurai map, illustrative).